

ACADEMIC TRAINING PROJECT

# QuickCafe Management System

Full-Stack QR Ordering, Table Routing, and Real-Time Kitchen  
Dashboard UI

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**Developer:** Training  
Student

**Database:** MongoDB Atlas  
(Cloud)

**Server:** Node.js +  
Express

**Theme Design:**  
Glassmorphism UI

# 1. Introduction & Objectives

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## What is QuickCafe?

QuickCafe is an interactive web-based dining application designed to digitalize café workflows. It bridges the gap between customer tables and kitchen staff, automating everything from digital menu browsing to order status tracking and billing.

## Core Project Objectives

- **Digitalize Customer Ordering:** Eliminate paper menus and physical order books using Table-specific QR Codes.
- **Automate Kitchen Workflows:** Real-time admin queue tracking orders, preparing stages, and kitchen handoffs.
- **Reduce Hardware Dependencies:** Run the entire dashboard, database, and billing terminal directly via a web server without heavy local server setups.
- **Deliver Premium Visuals:** High-end, responsive design built with Frosted Glassmorphism, animated lighting effects, and dual theme support.

## 2. Full-Stack Tech Stack & Database

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### Backend & Server Layer

**Node.js & Express:** The engine handling routing, API responses, tax calculations, and database connection pools.

Node.js

Express.js

REST APIs

Session Cookie Auth

### Database & Storage Layer

**MongoDB Atlas:** Multi-tenant Cloud Database storing collections for items, active categories, tables, active sessions, and orders.

MongoDB Atlas

NoSQL Schema

Counters Indexing

Browser LocalStorage

### Frontend Presentation Layer

**Pure Vanilla Stack:** Built on HTML5 and custom CSS3 to execute cutting-edge visual properties (floating backdrop mesh orbs, scale/shadow keyframes, scroll snaps) paired with vanilla JS DOM rendering.

- No heavy compilation needed (runs instantly on standard browsers).
- Robust client-side localStorage cart stores selected menu item IDs.
- Device responsive layouts (adjusts smoothly to mobile, tablet, and desktop).

### 3. Customer Flow & UI Experience

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#### Interactive Browsing Features

##### QR Table Detection

Scanning the table QR code auto-assigns table metadata. E.g., loading `table.html?table=12` configures the customer cart and routes it automatically to Table 12 on the admin side.

##### Popular Picks Spotlight Carousel

Highlights hot-selling drinks and snacks automatically. An animated spotlight cycles through the items, auto-scrolling horizontal cards cleanly into view.

##### Promo Code Engine

Supports validation codes: `WELCOME10` (10% off) or `FLAT20` (20% off for carts over ₹599) with live feedback.

#### The Checkout & Order Status Cycle

- **Select & Customize:** Add food items, adjust quantities with compact steppers, and view live totals.
- **Method Choice:** Checkout page asks for payment mode (UPI, Cash, Card) and triggers order submission.
- **Live Tracking Progress:** Submitting redirects the user to `status.html`, querying order documents (Pending → Preparing → Ready → Served) with a visual status bar.

## 4. Admin & Kitchen Management

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### Secure Staff Operations

#### Admin Authentication

Secured via bcryptjs encrypted password verification and active Express server cookies ( `qc_admin_session` ) to block unauthorized table manipulations.

#### Active Order Board

Categorizes orders dynamically. Admin can mark orders as *Preparing*, *Ready*, or *Served*. Status changes sync to customer tracking screens instantly.

#### Digital Menu Manager

Allows adding new items, editing prices, descriptions, and toggling dish availability ("Available" / "Sold Out") in a single click.

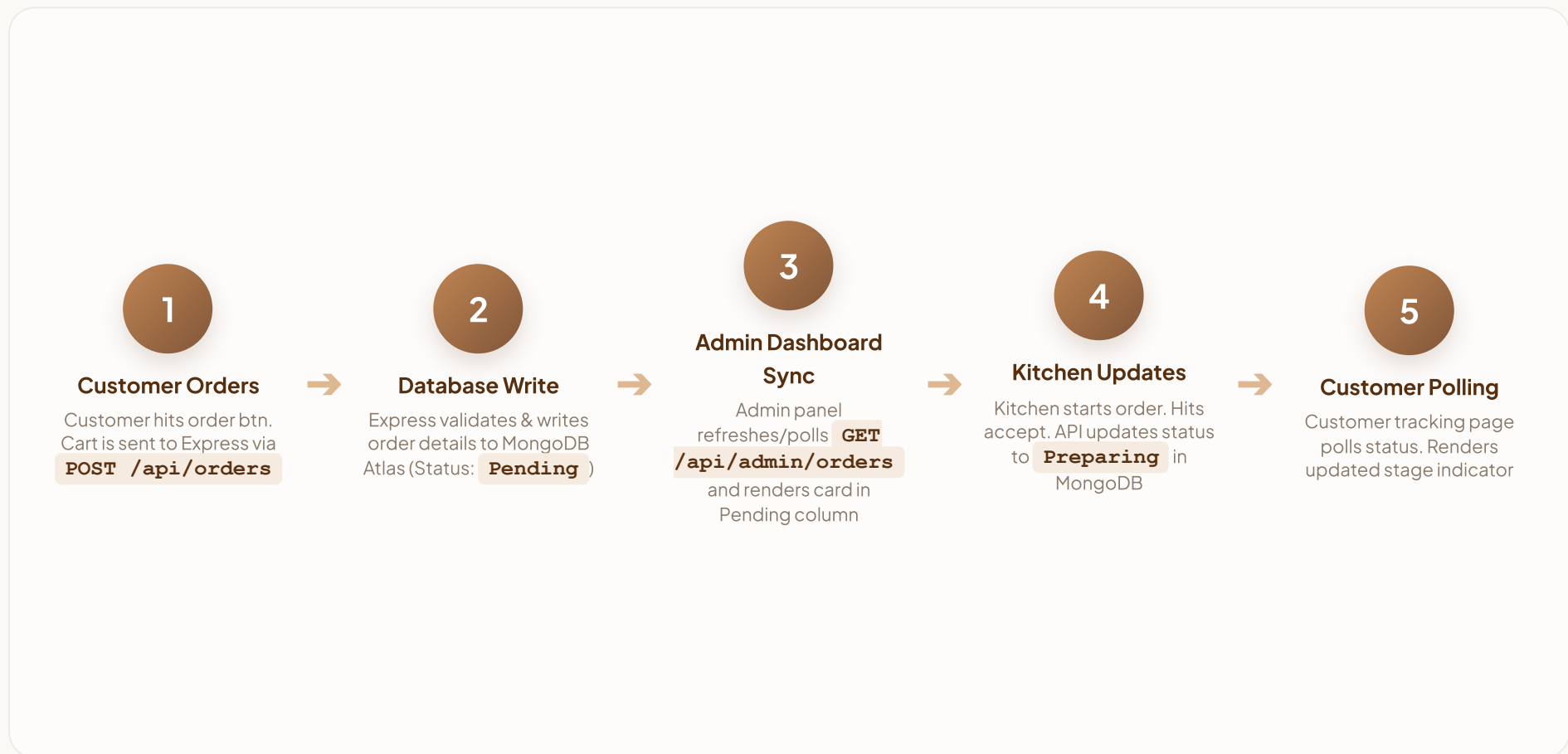
### Table QR Builder & Billing Terminal

- **QR Code Generator:** Admin can view and generate table-specific QR links directly, downloading print-ready QR codes for table placements.
- **Print-Friendly Billing:** Generates automated PDF invoices upon checkout, calculating tax (5% GST), applying promo discounts, and formatting a clean, receipt-printer receipt.

## 5. The Communication Bridge Explained

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Communication between Customer & Admin dashboards is completely decoupled and synchronized via MongoDB and REST API endpoints:



## 6. Premium Glassmorphic Design Aesthetics

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### Dual Theme Aesthetics

#### Dark Coffee Mode

Rich roasted colors, warm espresso gradients, and copper active highlights. Evokes a cozy, evening café vibe.

#### Light Eco Café Mode

Warm beige bases, soft white card panels, and organic green highlight accents. Delivers a bright, modern eco café feel.

### Glassmorphism UI Specifications

- **Frosted Refraction:** Card containers use translucent color base variables ( `rgba(255,255,255,0.50)` in light mode) blended with `backdrop-filter: blur(25px)`.
- **Ambient Depth:** Built with two massive, background glowing mesh orbs (gold/green/amber) drifting continuously behind the shell via asynchronous CSS keyframe coordinates.
- **Interactive Micro-Animations:** Smooth scaling transformations, color blends, and shadow transitions on interactive pills and add buttons.